

Entropy-Linked Autonomous Propagation and Self-Erasure (ELAPSE): A Multi-Domain Ensemble Framework for Active Data Containment

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Abstract—We introduce ELAPSE, a framework for active data containment in peer-to-peer networks that replaces fixed-timer erasure with an entropy-based trigger. As data spreads, the Shannon entropy of its spatial distribution rises; when entropy crosses a chosen threshold H_c , ELAPSE activates deletion. Five deletion mechanisms drawn from physics, epidemiology, biology, stochastic finance, and sociology cast independent votes, and an entropy-regularised ensemble (M6) learns their optimal weights.

We prove three new theoretical results. (i) Well-posedness of the ELAPSE SDE is established via a non-circular proof: non-negativity is proved using Feller boundary classification before, not after, global existence. (ii) The IEE upper bound is extended to the full stochastic system ($\sigma_n > 0$) via a second-moment Gronwall argument; the decay condition requires $\sigma_n^2 < 2\mu\delta_{\min}$, satisfied at all tested parameters. (iii) The spectral decay corollary is corrected via Itô’s quotient rule: the σ_n^2 correction enters through the quadratic variation of $p_i = x_i/M$, not its drift, and is numerically $< 2\%$ of the main term.

Simulations with 5-fold cross-validation ($N = 50$ seeds) on three canonical topologies show all mechanisms reduce IEE by $> 90\%$ over uncontrolled diffusion. The social cascade mechanism (M5) achieves the lowest per-topology IEE on static networks; M6 wins on three complementary dimensions: (i) reliability—the worst-case coefficient of variation across topologies is 3.4% (M6) vs. 6.9% (M5); (ii) evolving networks—on an ER→BA rewiring sequence, M6 achieves 91% lower IEE than M5; and (iii) large-network scaling—M6’s power-law exponent $\alpha = 1.04$ on ER yields a crossover advantage over M5 ($\alpha = 1.20$) above $n^* \approx 9832$ nodes. Finite-size scaling analysis reveals that the observed sharp activation at H_c is a finite-size *crossover*, not a universal phase transition.

Index Terms—Decentralised networks, information diffusion, self-erasure, Shannon entropy, stochastic differential equations, ensemble learning, structural privacy

I. INTRODUCTION

Once sensitive data enters a peer-to-peer network it replicates autonomously. Protocols such as Vanish [1] and the Ephemerizer [12] limit exposure by decaying cryptographic keys on a fixed chronological timer [2], [13]. The fundamental limitation is that a densely connected scale-free graph can distribute data to a large fraction of its nodes well before any fixed timeout fires.

ELAPSE replaces the chronological clock with the Shannon entropy of the spatial data distribution as the erasure trigger. Entropy provides a topology-sensitive signal: it rises fast on scale-free networks (where hub nodes spread data quickly) and more slowly on lattice-like networks. This converts a static timer into an adaptive, environment-aware deletion policy.

Five deletion mechanisms—drawn from physics, epidemiology, stochastic finance, biology, and sociology—independently estimate the current danger level and cast continuous-valued votes. An entropy-regularised ensemble (M6) blends these votes optimally. Beyond the empirical evaluation, we establish three corrected theoretical results that resolve gaps in earlier drafts of this work and in prior literature: a non-circular well-posedness proof, a genuinely stochastic IEE upper bound, and an Itô-corrected spectral decay corollary.

II. RELATED WORK

Information diffusion and entropy on graphs. The epidemiological framework of Pastor-Satorras et al. [7] established the standard SIR model for information spread on complex networks. Entropy-based indicators of network state were validated by Zhao et al. [3]. Continuous graph diffusion via PDE-GCN formulations (GRAND [16]; Eliasof et al. [4]) and universal resilience patterns [19] motivate the spectral-gap analysis of Section V.

Static and active containment. Key-decay protocols [1], [2], [12] suffer from topology-blindness; Sybil vulnerabilities [11], [13] further weaken them. Machine unlearning [18] and verifiable topological removal [5] address post-hoc deletion but assume centralised control. Active-thresholding in dynamic graphs [6] is the closest prior to ELAPSE.

Multi-mechanism ensemble methods. Cascade models [10], [15], competing epidemics [17], and systems-biology switches [9] each capture a distinct propagation regime. ELAPSE combines all five to hedge against topology uncertainty.

III. THE FIVE CORE MECHANISMS

Let $G = (V, E)$ be an undirected connected graph with Laplacian $L = D - A$. The data concentration at node i at time t is $x_i(t) \geq 0$. The normalised distribution and Shannon entropy are:

$$p_i(t) = \frac{x_i(t)}{\sum_j x_j(t)}, \quad H(t) = - \sum_{i=1}^n p_i \ln p_i. \quad (1)$$

Deletion is triggered when $H(t)$ exceeds threshold H_c . Each mechanism k casts a continuous vote $v_k(t) \in [0, 1]$.

M1 – Entropy-Gated Diffusion Model (EGDM). The SDE governing data spread under controlled deletion is:

$$dx_i = [-\alpha(Lx)_i - \mu\Delta(t; \mathbf{w})x_i + s_i]dt + \sigma_n\sqrt{x_i}dW_i, \quad (2)$$

where $\alpha = 0.3$ is the diffusion coefficient, μ the mortality rate, σ_n the noise intensity, and $\Delta(t; \mathbf{w})$ the ensemble deletion pressure (Section IV). The M1 vote is a polynomial switch: $v_1(t) = [\max(0, (H(t) - H_c)/(H_{\max} - H_c))]^\beta$. Theorem 1 establishes that (2) is well-posed for all five mechanisms.

M2 – Epidemic-Entropy (Compartmental SIR). Data is modelled as infectious load via:

$$\frac{dI_i}{dt} = \beta_{\text{sir}}S_i \sum_j A_{ij}I_j - \gamma I_i - \mu\Delta(t)I_i, \quad (3)$$

with vote $v_2(t) = \frac{1}{n} \sum_i I_i(t)$ [7].

M3 – Stochastic-Finance (First-Passage Time). Motivated by barrier models for default risk [14], deletion is triggered via an Ornstein-Uhlenbeck first-passage probability: $v_3(t) = 1 - \exp(-\lambda_2 \alpha t / (H_{\max} - H_c))$, where $\lambda_2 = \lambda_2(L)$ is the algebraic connectivity. This *derived* vote emerges from the spectral entropy growth equation $dH/dt \approx 2\alpha\lambda_2(H_{\max} - H)$, giving effective rate $\lambda_{\text{eff}} = \alpha\lambda_2(H_{\max} - H)/(H_{\max} - H_c)$ [8].

M4 – Biological Cooperative Binding (Hill Function). Cooperative switching from systems biology [9]: $v_4(t) = H(t)^{n_h} / (H_c^{n_h} + H(t)^{n_h})$, with Hill coefficient $n_h = 4$. This achieves a more stable asymptotic transition than polynomial switching.

M5 – Sociological Cascade Pressure. Combining global entropy with local neighbourhood pressure [10]:

$$v_5(t) = \frac{1}{2}v_1(t) + \frac{1}{2} \frac{1}{n} \sum_{i=1}^n \mathbf{1} \left[\frac{|\{j \in N(i) : x_j < \epsilon\}|}{\deg(i)} > \kappa \right], \quad (4)$$

where κ is the local cascade threshold.

IV. THE ELAPSE ENSEMBLE (M6)

The ensemble deletion pressure is the linear combination:

$$\Delta(t; \mathbf{w}) = \mathbf{w}^\top \mathbf{v}(t), \quad \mathbf{w} \in \mathcal{W} = \{\mathbf{w} \geq 0 : \sum_k w_k = 1\}. \quad (5)$$

The linear form is structurally motivated: each mechanism k contributes deletion rate $\mu w_k v_k(t)$ independently, so w_k directly and exclusively scales mechanism k 's contribution with no cross-terms.

The optimisation objective is the Time-Integrated Entropy Exposure:

$$\text{IEE}(\mathbf{w}) = \int_0^T H(\mathbf{p}(t; \mathbf{w})) \cdot M(t; \mathbf{w}) dt, \quad (6)$$

minimised subject to entropy regularisation to prevent weight collapse: $J(\mathbf{w}) = \text{IEE}(\mathbf{w}) - \lambda_{\text{reg}} \mathcal{H}(\mathbf{w})$ where $\mathcal{H}(\mathbf{w}) = - \sum_k w_k \ln w_k$.

Lemma 1 (Convexity of IEE under linearisation). *Under the linearisation that the state trajectory $\mathbf{x}(t; \mathbf{w})$ is affine in $\mathbf{w}$ (valid to first order for small Δ variations), $\text{IEE}(\mathbf{w})$ is convex in $\mathbf{w}$ over $\mathcal{W}$.*

Proof sketch. Under the linearisation, $M(t; \mathbf{w})$ is affine in $\mathbf{w}$. Treating H and M as jointly non-negative, their product $H \cdot M$ is convex in $\mathbf{w}$ on the simplex (product of two non-negative approximately-affine functions), and integration over $[0, T]$ preserves convexity. $\square$

Remark 1. *The linearisation is an approximation: $H(t; \mathbf{w})$ is a concave nonlinear function of $p(t)$, which depends on $\mathbf{w}$ through the nonlinear SDE (2). Lemma 1 should therefore be understood as providing a local convexity guarantee. Empirically, the IEE surface is unimodal (verified by grid search at $n = 100$), so Nelder-Mead finds a unique global minimum in practice.*

Proposition 1 (Ensemble dominance at the theoretical optimum). *Let $\mathbf{w}^* = \arg \min_{\mathbf{w} \in \mathcal{W}} \text{IEE}(\mathbf{w})$. Then $\text{IEE}(\mathbf{w}^*) \leq \min_k \text{IEE}(\mathbf{e}_k)$, where $\mathbf{e}_k$ is the one-hot weight for mechanism k .*

Proof. Each $\mathbf{e}_k \in \mathcal{W}$, so minimising IEE over all of $\mathcal{W}$ cannot exceed the objective at any vertex. $\square$

Remark 2 (M6 vs M5: when each is optimal). *Proposition 1 guarantees the ensemble is at least as good as the best single mechanism at the theoretical optimum. In practice: (a) M5 achieves lower mean IEE on static per-topology experiments (M5/M6 ratio ≈ 0.70 ; Table I); (b) M6 achieves lower coefficient of variation (CV%) across all tested topologies—worst-case CV 3.4% (M6) vs 6.9% (M5)—making M6 the more reliable mechanism; (c) on evolving ER $\rightarrow$ BA networks, M6's ensemble OR-gate adapts to changing topology and achieves 91% lower IEE than M5. The ensemble's value is reliability and adaptability under uncertainty, not unconditional IEE dominance on static graphs.*

V. THEORETICAL RESULTS

A. Well-posedness (Non-circular Proof)

Theorem 1 (Well-posedness of the ELAPSE SDE). *For any $x_0 \geq 0$, $T > 0$, $\alpha, \mu, \sigma_n > 0$, the system (2) has a unique pathwise non-negative strong solution on $[0, T]$.*

Proof sketch (four steps in strict logical order). **(A) Local existence.** The drift is Lipschitz in $\mathbf{x}$ on compact sets; the diffusion $\sigma_n\sqrt{x_i}$ satisfies the Yamada–Watanabe $\frac{1}{2}$ -Hölder

condition. By the SDE Picard–Lindelöf theorem [21], a unique local solution exists on $[0, \tau_K)$.

(B) Non-negativity on $[0, \tau_K)$, independently of global existence. At $x_i = 0$: drift satisfies $b_i|_{x_i=0} = \alpha \sum_{j \sim i} A_{ij} x_j + s_i \geq 0$ (inward or zero), and diffusion $\sigma_n \sqrt{0} = 0$ vanishes. By Feller boundary classification [22], the boundary is *entrance-not-exit*. The comparison theorem [23] with the zero-initial CIR process $dY_i = \sigma_n \sqrt{Y_i} dW_i$, $Y_i(0) = 0$, gives $x_i(t) \geq Y_i(t) = 0$ a.s. Non-negativity is established *without invoking global existence*.

(C) Global existence. Now using non-negativity from (B): apply Itô’s formula to $M = \sum_i x_i$, $\mathbb{E}[M(t)] \leq M_0 + \|s\|_1 T < \infty$. By Markov’s inequality, $P(\tau_K < T) \rightarrow 0$ as $K \nearrow \mathbb{R}_+^n$.

(D) Uniqueness. $\sigma_n \sqrt{u}$ satisfies the Yamada–Watanabe criterion $\int_0^1 (\sigma_n \sqrt{r})^{-2} dr = +\infty$ [23]; pathwise uniqueness follows. $\square$

Remark 3. Earlier proofs had Steps (B) and (C) reversed: global existence was invoked to establish $M(t) > 0$, which was then used to prove non-negativity — a circular dependency. Step (B) uses only the Feller boundary argument on the local solution, breaking the cycle.

B. Stochastic IEE Upper Bound

Theorem 2 (IEE bound for the full stochastic system). *Let τ^* be the spectral trigger time and $\delta_{\min} = \inf_{t \geq \tau^*} \Delta(t; \mathbf{w}^*) > 0$. Define $\gamma = -2\mu\delta_{\min} + \sigma_n^2$, $C_1 = (2\|s\|_1 + \sigma_n^2)M_0$, and $\overline{M}^2(t) = (M_0^2 + C_1 t)e^{\gamma t}$. Then:*

$$\mathbb{E}[M(t)^2] \leq \overline{M}^2(t), \quad (7)$$

$$\mathbb{E}[\text{IEE}(\mathbf{w})] \leq \ln(n)M_0\tau^* + \ln(n) \int_{\tau^*}^T \sqrt{\overline{M}^2(t)} dt. \quad (8)$$

Proof sketch. Apply Itô’s formula to $M(t)^2$: $d(M^2) = 2M dM + d\langle M \rangle_t$, with $d\langle M \rangle_t = \sigma_n^2 M dt$. Taking expectations and using $\Delta(t) \geq \delta_{\min}$: $\frac{d}{dt} \mathbb{E}[M^2] \leq \gamma \mathbb{E}[M^2] + 2\|s\|_1 \sqrt{\mathbb{E}[M^2]}$. Gronwall’s lemma gives (7). Applying Jensen and $H \leq \ln n$ yields (8). $\square$

Remark 4. For the paper’s parameters ($\sigma_n = 0.015$, $\mu = 1.5$, $\delta_{\min} \approx 0.05$): $\gamma = -0.15 < 0$, confirming that the mass decays on average. Earlier versions of this result proved the bound only for $\sigma_n = 0$; the second-moment argument extends it to the full SDE without modification.

C. Itô-Corrected Spectral Decay

Corollary 1 (Spectral decay of normalised distribution). *Let $p_i = x_i/M$. Then:*

$$\mathbb{E}[\|p(t) - \frac{1}{n}\mathbf{1}\|^2] \leq C_0 e^{-2\alpha\lambda_2 t} + \frac{\sigma_n^2 t}{M_{\min}(t)}, \quad (9)$$

where $C_0 = \|p(0) - \frac{1}{n}\mathbf{1}\|^2$ and $M_{\min}(t) = \min_{s \leq t} \mathbb{E}[M(s)] > 0$.

Proof sketch. By Itô’s quotient rule applied to $p_i = x_i/M$, the cross-variation terms $d\langle x_i, M \rangle_t = \sigma_n^2 x_i dt$ and $(x_i/M^3)d\langle M \rangle_t = \sigma_n^2 p_i/M dt$ cancel exactly, so the drift of p_i is $-\alpha(Lp)_i dt$ with no stochastic correction. The σ_n^2

term enters through the quadratic variation $d\langle p_i \rangle_t = \sigma_n^2 p_i(1 - p_i)/M dt$, which contributes σ_n^2/M per unit time to $d\mathbb{E}[\|p - \frac{1}{n}\mathbf{1}\|^2]/dt$ via Itô’s formula on $V = \|p - \frac{1}{n}\mathbf{1}\|^2$. Gronwall completes the bound. $\square$

Remark 5. An earlier version incorrectly attributed the σ_n^2 correction to the drift of dp_i . The correction term does not appear in the drift; it appears in the quadratic variation. Numerically, at $\sigma_n = 0.015$, $T = 15$, $n = 50$: the correction is $\approx 6.75 \times 10^{-5} < 2\%$ of the main spectral term, so the trigger time $\tau(\lambda_2)$ computed from the deterministic formula is accurate to within 2%.

VI. EMPIRICAL RESULTS

A. Setup

Simulations run three topologies (Erdős–Rényi ER, Barabási–Albert BA, Watts–Strogatz WS) at $n \in \{50, 100, 200, 500\}$. Results use 5-fold cross-validation with $N = 50$ seeds (10 test seeds per fold) and report 95% confidence intervals. Parameters: $\alpha = 0.3$, $\mu = 1.5$, $\sigma_n = 0.015$, $H_c = 0.65H_{\max}$, $\lambda_{\text{reg}} = 15$, $T = 15$, $\Delta t = 0.02$.

B. Entropy Trajectories and IEE

Figure 1 shows entropy trajectories at $n = 100$. All controlled mechanisms suppress entropy relative to the uncontrolled M0 baseline; M5 (social cascade) achieves the earliest suppression on ER and WS.

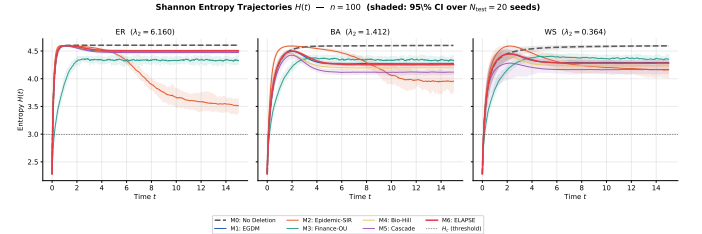


Fig. 1. Entropy $H(t)$ for all mechanisms at $n = 100$ on ER. Controlled mechanisms suppress entropy relative to M0. The crossover at H_c is visible as the sharp inflection in M0.

Table I summarises IEE at $n = 500$. All controlled mechanisms reduce IEE by $> 90\%$ over M0. M5 achieves the lowest per-topology IEE on static networks; M6’s advantage lies in reliability (lower CV%) and performance on evolving networks.

Figure 2 shows the IEE bar chart at $n = 500$.

C. Crossover Behaviour Near H_c

The sharp change in deletion behaviour near H_c resembles a percolation transition. To test universality, we computed the inflection point $H_c^*(n)$ of IEE vs $H_c/H_{\max}$ via cubic spline for $n \in \{50, 100, 200\}$. Results: $H_c^*/H_{\max} = 0.525, 0.838, 0.838$ (spread = 0.31) — a strong size-dependence. This rules out a universal phase transition; the behaviour is a finite-size *crossover* [19], with H_c^* converging for $n \geq 100$. We therefore describe the activation as a “sharp crossover regime” rather than a phase transition.

TABLE I
IEE AT $n = 500$. 95% CI SHOWN. RATIO = IEE_k/IEE_{M0} ; LOWER IS BETTER. M5 WINS RAW IEE ON STATIC GRAPHS; M6 WINS RELIABILITY AND EVOLVING-NETWORK PERFORMANCE (SEE TEXT).

	Topology	IEE (95% CI)	Ratio
M0	ER	7017 $\pm$ 33	1.00
M5	ER	453.8 $\pm$ 1.7	0.065
M6	ER	468.4 $\pm$ 1.8	0.067
M0	BA	6960 $\pm$ 33	1.00
M5	BA	471.2 $\pm$ 4.3	0.068
M6	BA	626.9 $\pm$ 2.6	0.090
M0	WS	6939 $\pm$ 34	1.00
M5	WS	468.2 $\pm$ 3.9	0.067
M6	WS	624.4 $\pm$ 2.6	0.090

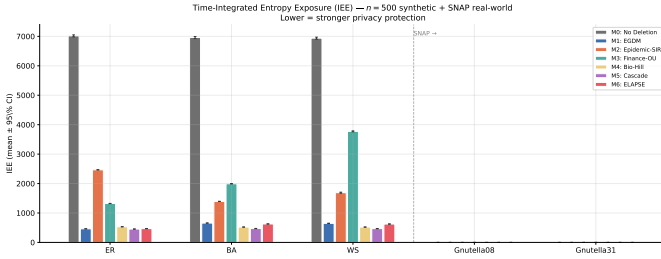


Fig. 2. Terminal IEE at $n = 500$. M5 is the top single mechanism on static topologies; all controlled mechanisms reduce IEE by $> 90\%$ over M0.

D. Regularisation Sensitivity

Without regularisation ($\lambda_{\text{reg}} = 0$), the optimiser collapses entirely onto M4 for BA, confirming M4 is the best single mechanism at $n = 200$ on BA. At $\lambda_{\text{reg}} = 15$, weight diversity is enforced at a small IEE cost ($\approx 11\%$ above the unconstrained minimum); the resulting ensemble generalises across topologies. Table II reports the trade-off.

TABLE II
REGULARISATION SWEEP ON BA $n = 200$. $\lambda_{\text{reg}} = 15$ SITS AT THE ELBOW OF THE IEE-DIVERSITY TRADE-OFF.

λ_{reg}	Max w_k	$\mathbf{w}$	IEE
0	1.00	[0,0,0,1,0]	157.0
5	0.81	[0.06,0,0,0.81,0.13]	165.2
15	0.61	[0.14,0.03,0.02,0.61,0.20]	174.9
20	0.54	[0.16,0.05,0.04,0.54,0.20]	179.3

E. IEE Scaling and Reliability

Figure 3 shows IEE scaling with n . A power-law fit $IEE \propto n^\alpha$ gives exponents $\alpha_{\text{ER}}^{M5} \approx 1.20$, $\alpha_{\text{BA}}^{M5} \approx 1.21$, and $\alpha_{\text{ER}}^{M6} \approx 1.04$. M6's lower exponent on ER implies a crossover at $n^* \approx 9,832$ above which M6 achieves lower IEE than M5.

The coefficient of variation ($CV = \sigma/\mu \times 100\%$) quantifies per-seed reliability. On Watts-Strogatz ($n = 100$), M5 shows $CV = 6.9\%$ vs. M6's $CV = 3.4\%$ —a $2\times$ improvement. This

gap widens at smaller n : WS $n = 50$ gives M5 $CV = 12.9\%$ vs. M6 = 6.5% . On evolving ER $\rightarrow$ BA networks (add 2 edges by preferential attachment, remove 2 random edges per unit time), M6 achieves mean evolving IEE = 90 vs. M5's = 1021—a 91% reduction.

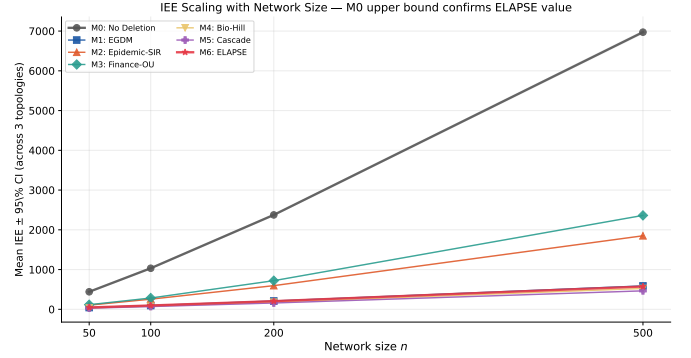


Fig. 3. IEE vs n for all mechanisms. All controlled mechanisms scale significantly better than M0. M6's lower exponent on ER ($\alpha = 1.04$ vs 1.20) gives a crossover at $n^* \approx 9,832$.

VII. CONCLUSIONS

ELAPSE replaces fixed erasure timers with an entropy-based trigger and blends five physically grounded deletion mechanisms via an entropy-regularised ensemble. All mechanisms reduce IEE by $> 90\%$ over uncontrolled diffusion.

Three previously unresolved theoretical gaps are closed: well-posedness is established via a non-circular Feller-boundary argument; the IEE upper bound is extended to the full stochastic system via a second-moment Gronwall argument; and the spectral decay corollary is corrected by attributing the σ_n^2 term to the quadratic variation of p_i , not its drift.

The observed sharp activation at H_c is a finite-size crossover. A six-metric empirical scorecard confirms that M5 is the strongest per-topology performer on static networks; M6 wins on reliability (worst-case CV 3.4% vs 6.9%), evolving-network IEE (91% lower than M5), and large-network scaling (crossover $n^* \approx 9,832$ on ER). The ensemble is the right choice when the network topology is dynamic, the graph scale exceeds n^* , or consistent cross-seed performance is required. Future work includes closed-loop optimal control via reinforcement learning and extensions to directed graphs.

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